

Sol. Let the numbers be $17a$ and $17b$.

Then, $17a \times 17b = 17 \times 714 \Rightarrow ab = 42$.

Now, co-primes with product 42 are (1, 42), (2, 21), (3, 14) and (6, 7).

So, the numbers are $(17 \times 1, 17 \times 42)$, $(17 \times 2, 17 \times 21)$, $(17 \times 3, 17 \times 14)$ and $(17 \times 6, 17 \times 7)$.

Since both the numbers are 3-digit numbers, the suitable pair is $(17 \times 6, 17 \times 7)$, i.e., (102, 119).

$\therefore$ Required sum = $102 + 119 = 221$.

Ex. 19. Find the least number which is exactly divisible by 32, 36, 45, 60 and 80.

(R.R.B., 2006)

Sol. Required number = L.C.M. of 32, 36, 45, 60, 80

2	32	–	36	–	45	–	60	–	80
2	16	–	18	–	45	–	30	–	40
2	8	–	9	–	45	–	15	–	20
3	4	–	9	–	45	–	15	–	10
5	4	–	3	–	15	–	5	–	10
3	4	–	3	–	3	–	1	–	2
2	4	–	1	–	1	–	1	–	2
	2	–	1	–	1	–	1	–	1

L.C.M. = $2 \times 2 \times 2 \times 3 \times 5 \times 3 \times 2 \times 2 = 1440$.

Hence, required number = 1440.

Ex. 20. Find the least number which when divided by 6, 7, 8, 9 and 12 leaves the same remainder 1 in each case.

Sol. Required number = (L.C.M. of 6, 7, 8, 9, 12) + 1.

3	6	–	7	–	8	–	9	–	12
2	2	–	7	–	8	–	3	–	4
2	1	–	7	–	4	–	3	–	2
	1	–	7	–	2	–	3	–	1

$\therefore$ L.C.M. = $3 \times 2 \times 2 \times 7 \times 2 \times 3 = 504$.

Hence, required number = $(504 + 1) = 505$.

Ex. 21. Find the smallest number which when increased by 10 is completely divisible by 12, 15, 18, 20 and 24.

(P.C.S., 2008)

Sol. Required number

= (L.C.M. of 12, 15, 18, 20, 24) – 10

= $(2 \times 2 \times 3 \times 5 \times 3 \times 2) - 10$

= $360 - 10 = 350$.

2	12	–	15	–	18	–	20	–	24
2	6	–	15	–	9	–	10	–	12
3	3	–	15	–	9	–	5	–	6
5	1	–	5	–	3	–	5	–	2
	1	–	1	–	3	–	1	–	2

Ex. 22. Find the greatest number of five digits which is divisible by 15, 21 and 36.

(P.C.S., 2010)

Sol. Greatest number of five digits = 99999.

Required number must be divisible by L.C.M. of 15, 21, 36, i.e., 1260.

On dividing 99999 by 1260, we get 459 as remainder.

$\therefore$ Required number = $(99999 - 459) = 99540$.

Ex. 23. Find the smallest number of five digits exactly divisible by 16, 24, 36 and 54.

Sol. Smallest number of five digits is 10000.

Required number must be divisible by L.C.M. of 16, 24, 36, 54, i.e., 432.

On dividing 10000 by 432, we get 64 as remainder.

$\therefore$ Required number = $10000 + (432 - 64) = 10368$.

Ex. 24. Find the largest number which when subtracted from 10000, the remainder is divisible by 32, 36, 48 and 50.

(R.R.B., 2006)

Sol. Required number

= $10000 - (\text{L.C.M. of } 32, 36, 48, 50)$

= $10000 - (2 \times 2 \times 3 \times 4 \times 3 \times 2 \times 5)$

= $10000 - 864 = 9136$.

2	32	–	36	–	48	–	54
2	16	–	18	–	24	–	27
3	8	–	9	–	12	–	27
4	8	–	3	–	4	–	9
3	2	–	3	–	1	–	9
	2	–	1	–	1	–	3